Here is the complete Assignment 1 execution sequence, covering all 17 checklist items in Ubuntu Terminal.

You can perform each command one by one and verify the output.

1\. Create directory structure

```bash
mkdir -p ~/mock_test
cd ~/mock_test
pwd
```

```bash
mkdir -p a b c
touch a/file1.txt a/file2.txt a/file3.txt
cp a/file1.txt a/file2.txt a/file3.txt b/
mv a/file3.txt c/
```

Install and display the tree:

```bash
sudo apt install tree -y
tree
```

```

### 2\. File permissions and ownership

Create file:

```bash
touch permission.txt
```

Symbolic permission:

```bash
chmod u=rwx,g=rx,o= permission.txt
ls -l permission.txt
```

Octal permission:

```bash
chmod 750 permission.txt
ls -l permission.txt
```
```

Change owner and group automatically to your current user and primary group:

```bash
sudo chown "$(whoami):$(id -gn)" permission.txt
```

Display final permissions:

```bash
ls -l permission.txt
```

Expected:

```text
-rwxr-x--- 1 username groupname 0 ...
```

3\. Process management

A. Display all processes

```bash
ps aux
```

```

B. Find the highest memory-consuming processes

```bash
ps aux --sort=-%mem | head
```

C. Check a process before terminating it

Replace PID with the actual process ID:

```bash
ps -p PID -o pid,user,comm,%mem
```

If it is a process you own and can safely close:

```bash
kill -15 PID
```

Verify:
```

```bash
ps -p PID
```

Do not terminate essential system processes.

4\. System monitoring

A. CPU and memory monitoring

```bash
top
```

Press `q` to exit.

B. Disk usage

```bash
df -h
```

C. RAM usage

```bash
free -h
```

D. Logged-in users

```bash
w
```

5\. Final verification

Run:

```bash
cd ~/mock_test
tree
ls -l permission.txt
ps aux --sort=-%mem | head
df -h
free -h
w
```

Complete checklist

Assignment 1

0/17

Create mock_test directory

Create directories a, b, c

Create three files inside a

Copy files from a to b

Move file3.txt to c

Display tree structure

Set symbolic permissions

Set octal permissions 750

Change owner and group

Verify using ls -l

Display processes using ps

Find highest memory process

Safely terminate selected process

Monitor CPU using top

Check disk using df -h

Check RAM using free -h

Display users using w

Reset checklist

Important: Complete the commands in order. Take screenshots of Q1, Q2 and Q3 outputs for your practical submission.